Eco Race

Turning Environmental Actions into Rewards and Competition in Italy

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Abstract

In today's world, environmental degradation has become one of the major threats due to industrial and factory activities [1],[9]. Traditional approaches, which rely only on warnings and rules, often have a temporary effect and do not lead to long-term changes. The main purpose of this project is to transform the old methods into a competitive system through the development of the application (Eco Race). This app links environmental protection to the "reward and competition" system (Gamification). The way the project works relies on providing a mixture of environmental tasks and challenges (such as planting trees) in which users accumulate points [4]. These points lead to a competitive leaderboard at the level of individuals and educational institutions (universities and schools). The system also includes artificial intelligence (AI) to recommend daily tasks and publish the latest environmental news [10]. A verification system based on Blockchain technology is also included to ensure that users complete real activities before receiving points [7]. This is done by allowing users to upload proof such as images or activity confirmations as shown in Fig. 4., which are checked before points are given. This helps reduce fake submissions and keeps the system fair for all users. Overall, this project aims to change the way people interact with the environment today by introducing gamification and competition, making environmental actions more engaging through rewards and points [5],[8].

Project Objectives

The goals of the Eco-Race project are:

- To design a mobile application that motivates users to take environmental action through competition and rewards.
- To create a credit-based system that allows users to earn credits for completing different eco-friendly tasks.
- To create a leaderboard that compares each user and student to others in the application.
- To partner with local businesses so users can redeem their credits for real discounts and rewards.
- To build a fair and reliable verification system so that every credit earned represents a genuine real-world action.
- To create competition between schools and universities based on total user contribution.

Problem Statement

The world is currently suffering from environmental problems that are a direct result of rapid industrial growth and the proliferation of factories. This development has caused pollution of the air and soil [2]. Despite the existence of environmental laws and guidelines, the main problem is that most people see environmental protection as a difficult or boring task. Environmental awareness often only takes the form of temporary warnings and publishments, making their impact short-lived and unable to become part of an individual's daily culture. There is a huge gap between environmental knowledge and environmental action. People know that environmental protection is important, but they do not have a strong incentive to do so on a sustainable basis. The old ways have failed to efficiently attract young people and students into this field. The Eco-Race project identifies the lack of motivation and the lack of competition as the main reasons for the slowdown in environmental activities. In addition, there are no clear approaches that allow individuals to track their environmental contributions or see the results of their actions over time. Existing approaches do not provide rewards or recognition, which reduces user interest and participation. There is also a lack of structured competition between individuals, schools, and universities. The problem lies not only in pollution, but also in outdated methods that have failed to turn environmental protection into a sustainable habit.

Methodology and Approach

The development of Eco-Race followed a research driven design process, moving from problem identification through system planning and interface design. The process began with a review of existing sustainability applications and gamification platforms [3]. Most current eco-apps focus on passive tracking or awareness, offering little in the way of social interaction or reward. According to research on behavior change, external rewards and social comparison are two of the best motivators for constant activity. It has been proven that gamification frameworks, especially those built around points, leaderboards, and reward systems, significantly increase interest and sustained participation in goal-oriented tasks [3],[5],[8]. The design choices for this project were based on these conclusions. As a result, Eco-Race was designed using a gamified framework in which users get points for participating in verified eco-friendly activities. To guarantee the validity of submitted activities, a verification system is formed, enabling users to offer evidence such as photos or activity confirmation before receiving points. Institutional competition is one of the system's main features. While public users engage individually or within community ranks, students contribute points to their school or university. On different leaderboards, schools and universities compete with one another, promoting both individual and group activities as shown in Fig. 5, 6. Furthermore, an AI-based feature that offers a dynamic environmental news feed and suggests eco-friendly actions was added to increase participation [10]. This supports continuous user interaction by suggesting tips and keeping users informed. At first, the system was structured using flowcharts as shown in Fig. 1. to show how users navigate the app, from registering to finishing tasks and earning rewards. The interface was then created in Figma as shown in Fig. 2. With a focus on making sustainable tasks feel effective, the end result is a working framework that illustrates how the app works and how users interact with it.

Results and Impact

The application is expected to significantly boost user participation in sustainable activities. Users are encouraged to engage in eco-friendly activities more regularly through the use of points, prizes, and competition. The application might have between 1,000 and 3,000 active users in an initial launch with 5–10 schools and institutions, each with an estimated 200–300 users. Between 1,000 and 9,000 recorded actions might be recorded each week if each user completes one to three tasks. Over time, these actions — as planting trees, saving water, and reducing waste — would result in noticeable environmental improvements. The introduction of competition between users and institutions would boost engagement even further because people will be able to contribute to both their own progress and that of their school or university. Features like progress tracking and task recommendations can also support continuous engagement. All things considered, the application has the capacity to transform environmental awareness into consistent action, resulting in a competitive and expandable beneficial impact within communities.

Innovation and Creativity

What makes this project unique is that it makes environmentalism more effective and more engaging for users. The application employs a system of rewards and competition rather than short-term campaigns and guidelines to motivate users to regularly take part in eco-friendly activities. A key feature is the competition between users and institutions. Users can collect points for themselves as well as for their institutions, creating a real competition between institutions. This is not only competition, but also adds a sense of teamwork and responsibility. Also, the app has a verification system to ensure actions are correct, and this makes the system fair and credible. By adding suggestions of eco-friendly tasks and environmental news, the app protects user sustainability. Overall, this project is an attempt to make environmentalism powerful, interesting, and sustainable for a long period of time.

Scalability and Feasibility

The Eco-Race program is made to be flexible and expandable for practical use in a variety of environments. To assess performance and user engagement, the platform can first be deployed in a limited number of educational institutions. This makes it possible to test and improve before reaching more institutions and users. The system can be expanded to serve a wider network of participants by incorporating additional schools, institutions, and communities as acceptance rises. By allowing both individual users and institutions to communicate inside the same system, the application structure allows the system to adapt to different levels of use — meaning we can add more challenges, tasks, and collaborations. In terms of feasibility, the project relies on commonly used technologies such as mobile application frameworks and cloud-based storage, making it cost-effective and practical to implement. Long-term sustainability can be supported through collaboration with educational institutions and external organizations that provide incentives and rewards. Regular updates and competitions can also help maintain user engagement over time.

Visuals and Supporting Materials

The user flow chart is shown in Figure 1.

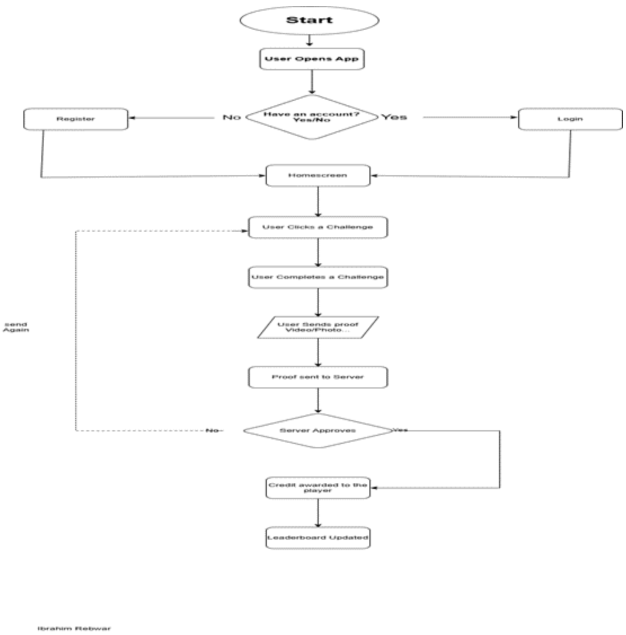

Figure 1: Eco Race User Flowchart

Application prototype:

The user interface of the application is shown in Figures 2–8.

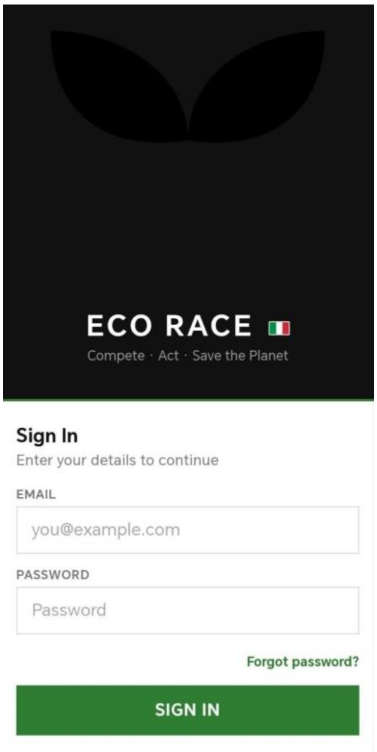

Fig.2. Sign In Menu

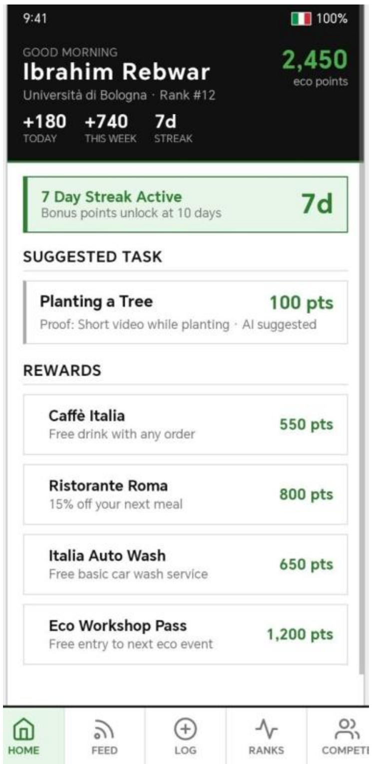

Fig. 3. Home Page

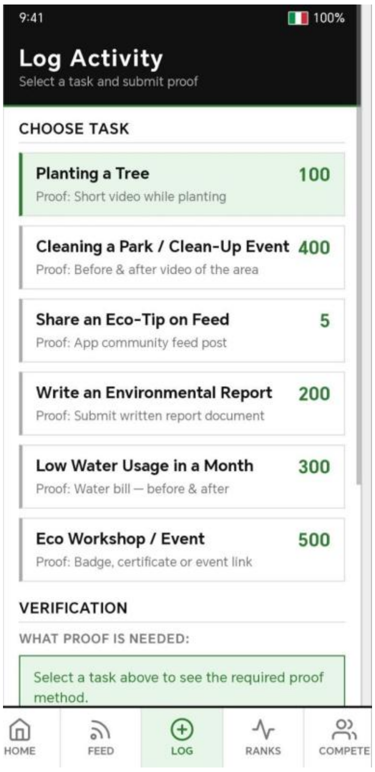

Fig. 4. Home Page

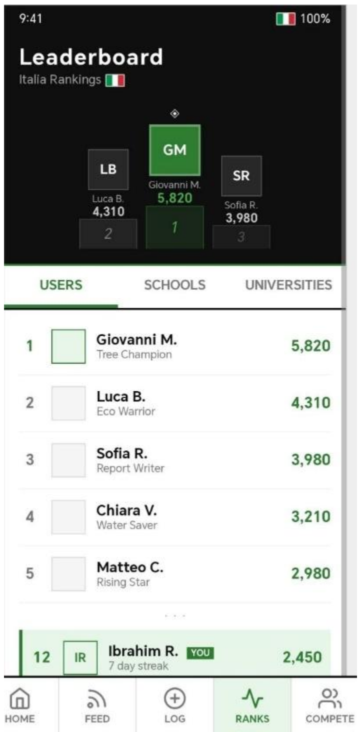

Fig. 5. Users leaderboard

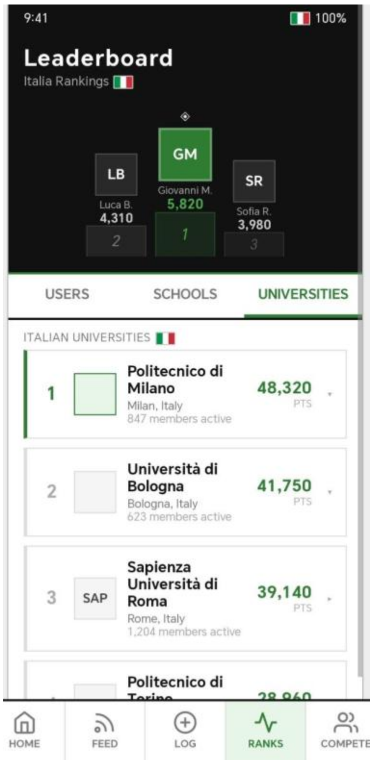

Fig. 6. Universities leaderboard

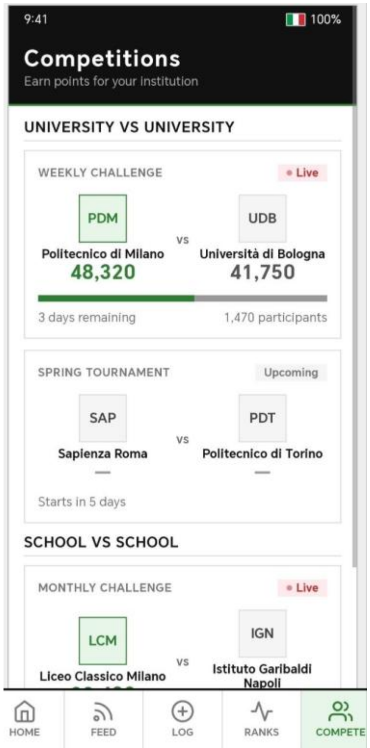

Fig. 7. Competitions Menu

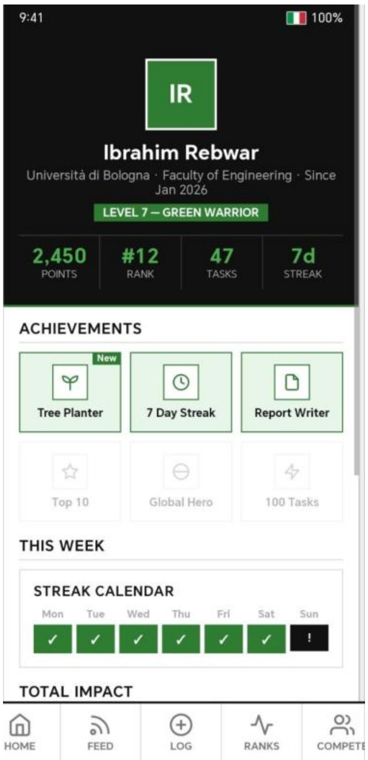

Fig. 8. User Profile

References and Sources

[1] E. Flanagan, "Industrial Pollution: The Growth of Environmental Degradation and Health Hazards," J. Pollut. Eff. Control, vol. 12, p. 392, Mar. 2024, doi: 10.35248/2375-4397.24.12.392.

[2] S. N. Shah, S. Manzoor, and M. Asim, "Impact of Industrial Pollution on Our Society," Pak. J. Sci., vol. 73, no. 1, pp. 1–8, Mar. 2021.

[3] Y. Chen, W. Liu, and X. Zhang, "Gamification Effects in Green Behaviors: A Double-Edged Sword," J. Clean. Prod., vol. 490, Nov. 2024, doi: 10.1016/j.jclepro.2024.037612.

[4] S. Lim, "Gamification for Sustainable Consumption: A State-of-the-Art Overview and Future Agenda," Bus. Strategy Environ., Nov. 2024, doi: 10.1002/bse.4021.

[5] A. Navarro-Espinosa et al., "Gamification for Sustainability: A Systematic Review of Applications, Trends, and Opportunities," Comput. Hum. Behav., vol. 165, p. 108529, 2024, doi: 10.1016/j.chb.2024.108529.

[6] A. Felfernig, M. Jeran, G. Ninaus, F. Reinfrank, and S. Reiterer, "Recommender Systems for Sustainability: Overview and Research Issues," Front. Big Data, vol. 6, Oct. 2023, doi: 10.3389/fdata.2023.1284511.

[7] L. Chen, H. Xu, and Z. Chen, "A Blockchain-based Crowdsourcing System for Large Scale Environmental Data Acquisition," in Proc. IEEE Int. Conf. Distributed Comput. Syst. (ICDCS), 2021, pp. 1–10, doi: 10.1109/ICDCS51616.2021.00050.

[8] A. Costa, M. Pereira, and R. Santos, "The Use of Gamification and Web-Based Apps for Sustainability Education," Sustainability, vol. 16, no. 8, p. 3197, Apr. 2024, doi: 10.3390/su16083197.

[9] M. Awais, T. Siddiqua, and S. Hasnain, "The Environmental Impact of Industrialization and Foreign Direct Investment: Empirical Evidence from Asia-Pacific Region," Environ. Sci. Pollut. Res., vol. 29, pp. 8439–8455, 2022, doi: 10.1007/s11356-021-17029-6.

[10] K. Al-Smadi, L. Barakat, and R. Hijazi, "A Novel Approach to Sustainable Behavior Enhancement through AI-Driven Carbon Footprint Assessment and Real-Time Analytics," Discov. Sustain., vol. 5, p. 762, Dec. 2024, doi: 10.1007/s43621-024-00762-w.